Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=3;i++)
   (i-1)[arr] = (i%3);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 1;
}
int main(){
  int arr[4] = {0,0,0,0};
  func1(arr+1);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output
0 1 2 0
1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

Yes
int temp is at random memory location, not necessarily cts to arr, so now a[0] will be the value at temp but a[1], a[2], a[3] will be garbage values.

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

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a_1 = address of 50th element of arr2, a_2 = address of 10th element
$= (50-10) \cdot 8$ (∵ long)
$b_1 - b_2 = (50-10) \cdot 4$ (∵ int)